

Multiple Choice (10 marks)

1. What is the approximate boiling point at standard pressure of a solution prepared by dissolving 234 g of NaCl in 500 g of H₂O?
 - (a) 102.5
 - (a) 100
 - (a) 109.5
 - (a) 106.78
 - (a) 101.22
2. A substance that is NOT generally considered to be a toxic pollutant in water is
 - (a) nitrogen
 - (b) a halogenated hydrocarbon
 - (c) chlorine
 - (d) copper
 - (e) arsenic
3. Of the following compounds, which has the lowest melting point?
 - (a) NaCl
 - (b) LiCl
 - (c) AlCl₃
 - (d) KCl
 - (e) FeCl₃
4. 10.0 grams of urea, CO(NH₂)₂, is dissolved in 200 grams of water. What are the boiling and freezing points of the resulting solution?
 - (a) Boiling point is 101.00°Freezing point is 1.55
 - (a) Boiling point is 101.00°Freezing point is - 2.00
 - (a) Boiling point is 99.57°Freezing point is 1.55
 - (a) Boiling point is 100.0°Freezing point is 0.0
 - (a) Boiling point is 100.43°Freezing point is 1.55

5. Calculate the weight in grams of sulfuric acid in 2.00 liters of 0.100 molar solution. (MW of $\text{H}_2\text{SO}_4 = 98.1$.)
- (a) 10.0 g
 - (b) 24.4 g
 - (c) 15.2 g
 - (d) 22.3 g
 - (e) 19.6 g

True / False (10 marks)

Answer True or False

- 6. True or False: Compounds like Na_2O and CaCl_2 also require Roman numerals in their names due to their ionic nature.
- 7. True or False: Diluting 50 ml of 3.50 M H_2SO_4 to achieve a concentration of 2.00 M requires a final volume of 125 ml.
- 8. True or False: The quotient $[\{2x^0\} / \{(2x)^0\}]$ simplifies to 1, since both the numerator and denominator evaluate to the same value.
- 9. True or False: To produce 1.50 g of O_2 , 8.45 g of NaNO_3 must be decomposed according to the reaction.
- 10. True or False: The zero-point entropy of NO is $0.00 \text{ cal mol}^{-1} \text{K}^{-1}$, indicating no disorder at absolute zero.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Calculate the cell potential E for the $\text{Ag}/\text{AgCl}/\text{Cl}^- (a = 0.0001)$ and $\text{Ce}^{4+} (a = 0.01), \text{Ce}^{3+} (a = 0.1)/\text{Pt}$ system, discussing the significance of standard half-cell potentials in your analysis.
- 12. Discuss the stoichiometric principles involved in the decomposition of mercury(II) oxide (HgO) to form oxygen gas (O_2) and liquid mercury (Hg). Calculate the moles of oxygen produced from 4.32 g of HgO , given its molar mass is 216 g/mol.

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